Group Project - StudyHub

Requirement and Design Specification

Contents

I. Overall Requirements 2

1. Screen Flow 2
2. Screen Descriptions 2
4. Screen Authorization 4
5. Main Workflows 5

II. Screen Layouts 6

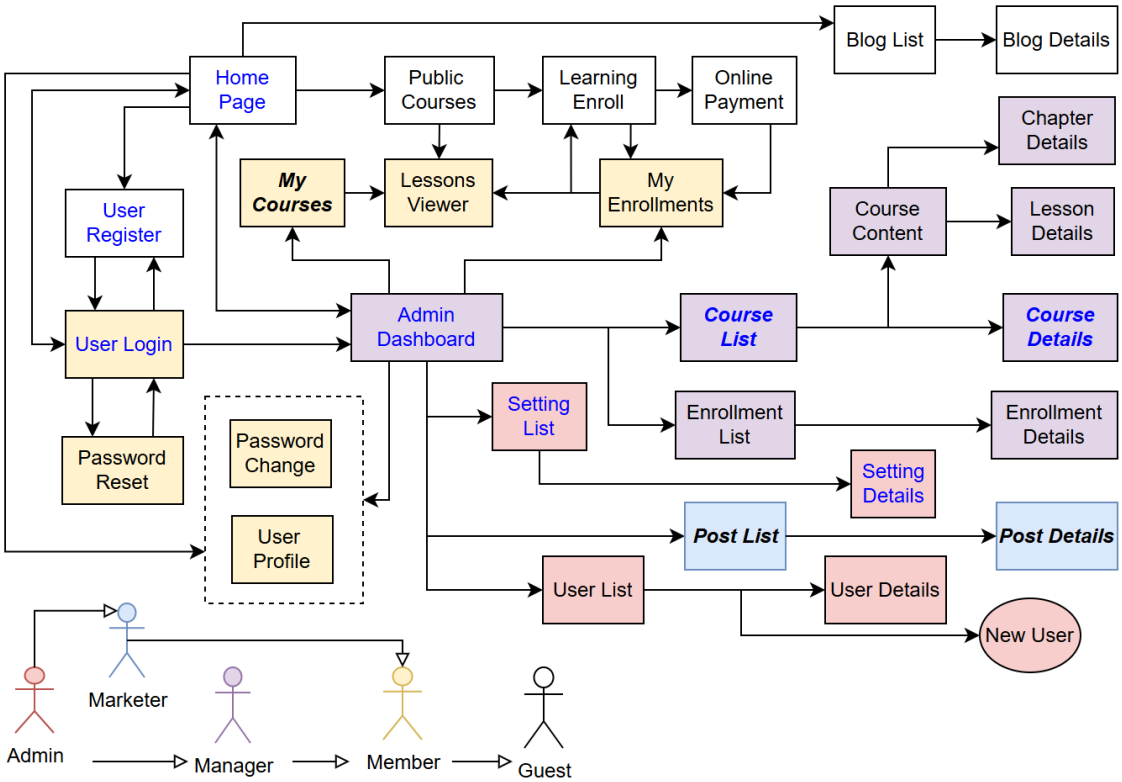
1. Common Feature 6
2. User Authentication 9
3. System Administration 10
4. Course Management 14
5. Course Enrollment 16
6. Course Access 18

III. System Design 19

1. High Level Table Design 19
2. System Architecture 19

I. Overall Requirements

1. Screen Flow



2. Screen Descriptions

Screen	Description
Home Page	Show featured (hot) active courses
Public Courses	Show all active courses, sorted by course name
User Register	Allow new user to register Member account to access the system; After registering, user need to verify his/her email by clicking the link in the account verifying email from the system. After clicking the verifying link, user is redirected to the User Login page with appropriate toast message
User Login	Authenticate user and give him/her accessing system's functionalities; Inactive and unverified user is not allowed to login. For unverified user, he/she would have option to request new account verifying email to verify his/her account
Password Reset	Allow user to reset his/her password for the case he/she forgot it; After user inputs valid email address, system would send user an email with the password-resetting link from that he/she can access the page for inputting new password; The password-resetting link is expired after 30 minutes.
User Profile	User can view and update his/her profile details
Password Change	Allow user to change his/her password; He/she would be logged out and redirected to the User Login page after successfully changing the password
Admin Dashboard	Admin and Manager can view full system statistics and latest courses. Admin can have the links to the Course Details and Course Content pages in all latest courses. However, with Manager role, those links are limitted to his/her assigned courses.
Setting List	View and activate/deactivate the system settings
Setting Details	Add new or update a specific system settings. The setting name must be unique within each group of system settings with the same setting type.
User List	View, delete or activate/deactivate the user accounts. Only new user (who has not had any transactions yet in the system) can be deleted

New User	Create new user, then generate a random password and send to newly-created user's email along with other account-creation notification
User Details	View and/or update details of a specific user account
Course List	View, delete or publish/unpublish the courses. Only new course (which are note related to any other system transactions in the system) can be deleted
Course Details	View, add new, or update course details. Manager can have accesses only to his/her assigned courses and he/she are not allowed to change the course price and status (published, unpublished)
Course Content	View details of the course and course contents (chapters, lessons); Manager has only accesses to the courses that he/she has been assigned; On this page, user can choose to delete a specific chapter or lesson or have links/buttons to access the Chapter Details and Lesson Details pages for adding new or updating details of specific chapter, lesson.
Chapter Details	Allow user to provide details for adding new or updating existing chapter; Manager has only accesses to the chapters of the courses that he/she has been assigned;
Lesson Details	Allow user to provide details for adding new or updating existing lesson; Manager has only accesses to the lessón of the courses that he/she has been assigned;
Learning Enroll	Both Guest and Member can enroll a specific course (if he/she has not been approved to join that course before). For the case of Member, his/her profile information would be automatically filled into the registration fields, and he/she can update to enroll the course for other people if neeeed; After successfull registered, a notification email would be sent to both registering and registered people informing about the success registration; If registered user does not exist, he/she would be registered into the system with newly-random access password and that password is sent to the registered user (along with the registering notification email)
Online Payment	This is for user to handle the bank transfering (using PayOS feature) or internet banking (using VnPay feature) payment according to the payment method that he/she has chosen from the previous step (Learning Enroll screen) and update to the enrollment status as approved for success payment
My Enrollments	Allow user to view his/her enrollments; For the approved enrollments, user can choose to view the enrollment details and registerd user's learning progress in a modal dialog. For the pending or rejected enrollments, user can choose to go back to the Learning Enroll screen to view rejection notes (if any), then updating and making payment.
Enrollment List	View, export the list of enrollments for the courses and import Excel enrollment list for a specific course; The user can also delete his/her created enrollments and approve or reject a specific enrollment (with relevant notes) The Manager can have accesses to enrollments of his/her assigned courses only while the Admin can have accesses to enrollments of all the courses
Enrollment Details	View, add new or update details for a specific enrollments; The Manager can have accesses to enrollments of his/her assigned courses only while the Admin can have accesses to enrollments of all the courses
My Courses	Allow user to view the courses that he/she has been approved to join
Lesson Viewer	User can view all the active chapters and lessons of the selected course; The chapters and lessons in each chapter are sorted in the way Admin or Manager has defined before. Each lesson is shown in the suitable form (video player, PDF viewer, or rich-text post) depending on its content type.
Post List	View (include filter, search, sort) or publish/hide the posts.
Post Details	View, add new, or update post details. Marketer can have accesses only to his/her owned post and he/she is not allowed to change the post status (published, hidden)
Blog List	View (include search by title and filter by category) all the published posts
Blog Details	View details of a chosen published post and related discussions (comments, replies) Member can add new or delete comment/reply to a specific comment or reply.

4. Screen Authorization

System provide the functionalities or accesses to 4 different user classes as listed in the table below. Details are provided in the description for each screen in the previous section.

Function	Admin	Manager	Marketing	Member	Guest
Home Page	x	x	x	x	x
Public Courses	x	x	x	x	x
Blog List	x	x	x	x	x
Blog Details	x	x	x	x	x
User Register					x
User Login	x	x		x	
Password Reset	x	x		x	
User Profile	x	x		x	
Password Change	x	x		x	
Admin Dashboard	x	x			
Setting List	x				
Setting Details	x				
User List	x				
New User	x				
User Details	x				
Post List	x		x		
Post Details	x		x		
Course List	x	x			
Course Details	x				
Course Content	x	x			
Chapter Details	x	x			
Lesson Details	x	x			
Learning Enroll	x	x		x	x
Online Payment	x	x		x	x
My Enrollments	x	x		x	
Enrollment List	x	x			
Enrollment Details	x	x			
My Courses	x	x		x	
Lesson Viewer	x	x		x	

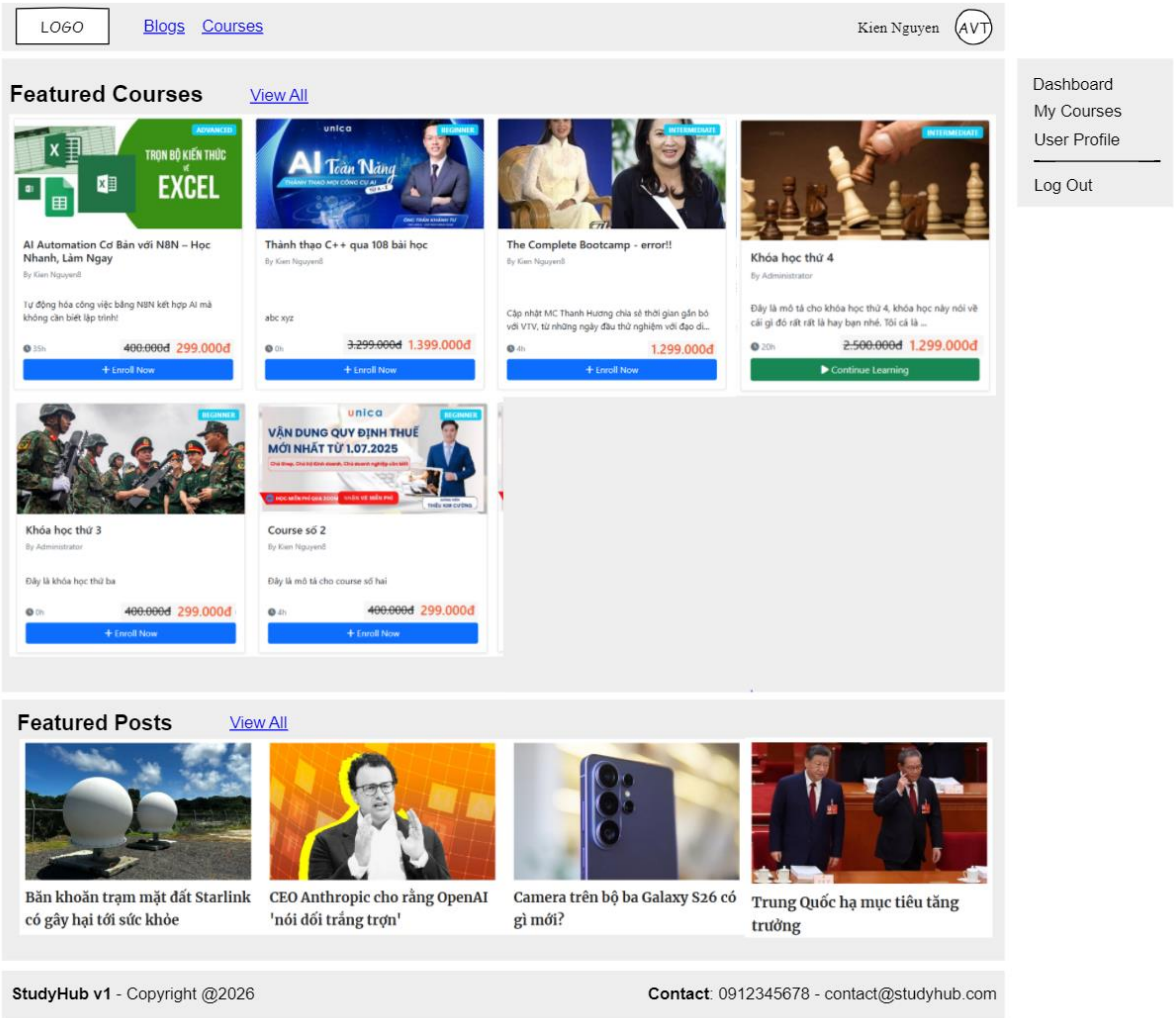
5. Main Workflows

Function	System Admin	User Authentication	Post Management	Course Management	Course Enrollment	Lesson Access
Home Page	x	x	x	x	x	x
Public Courses	x			x	x	
Blog List	x		x			
Blog Details	x		x			
User Register		x				
User Login	x	x		x		x
Password Reset		x				
User Profile	x	x				
Password Change		x				
Admin Dashboard			x	x		
Setting List	x					
Setting Details	x					
User List	x	x				
New User		x				
User Details	x	x				
Post List			x			
Post Details			x			
Course List	x			x		x
Course Details	x			x		x
Course Content						x
Chapter Details						x
Lesson Details						x
Learning Enroll					x	
Online Payment					x	
My Enrollments				x	x	
Enrollment List					x	
Enrollment Details					x	
My Courses				x	x	x
Lesson Viewer						x

II. Screen Layouts

1. Common Feature

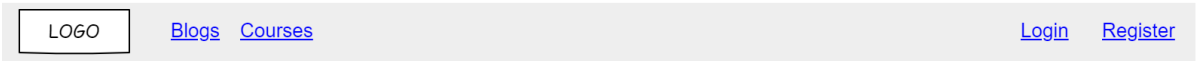
1.1. Home Page



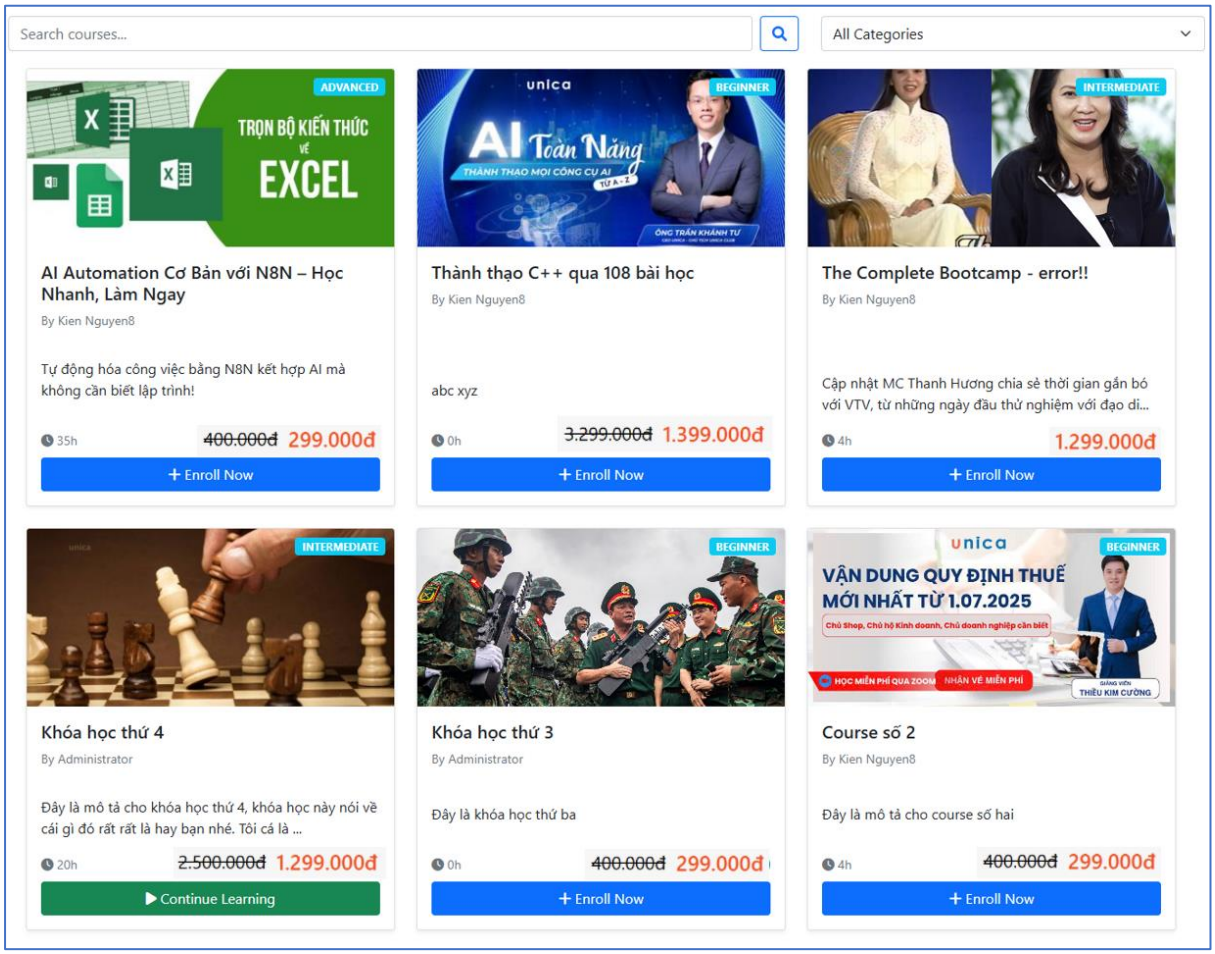
Logging-in users would see the “Continue Learning” button under the course(s) that he/she has paid.

The Dashboard menu on the Context menu (shown after clicking the user avatar) is shown only when logging-in user is Admin, Manager or Marketing, allowing them to access the Admin Dashboard page

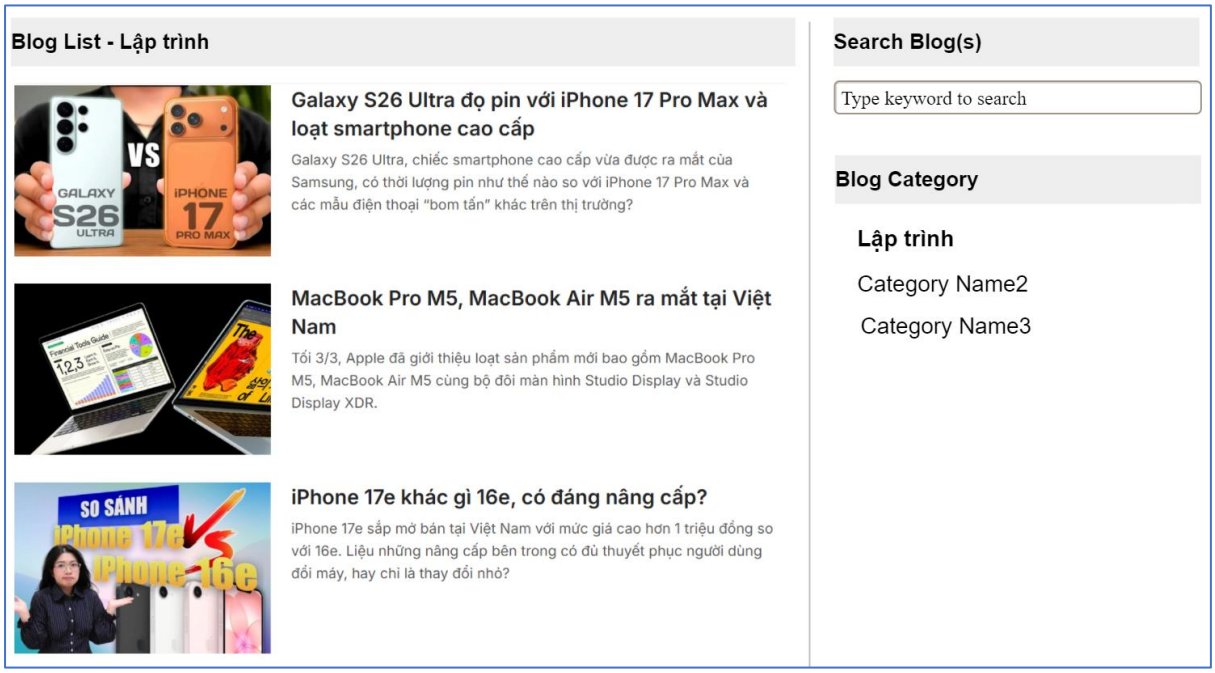
When the user has not logged in yet, he/she can see the “+ Enroll Now” button show only on each course and the two links/buttons Login and Register would be shown instead of the Avatar and user’s Full Name at the top-right corner for accessing to the User Login and User Register pages.



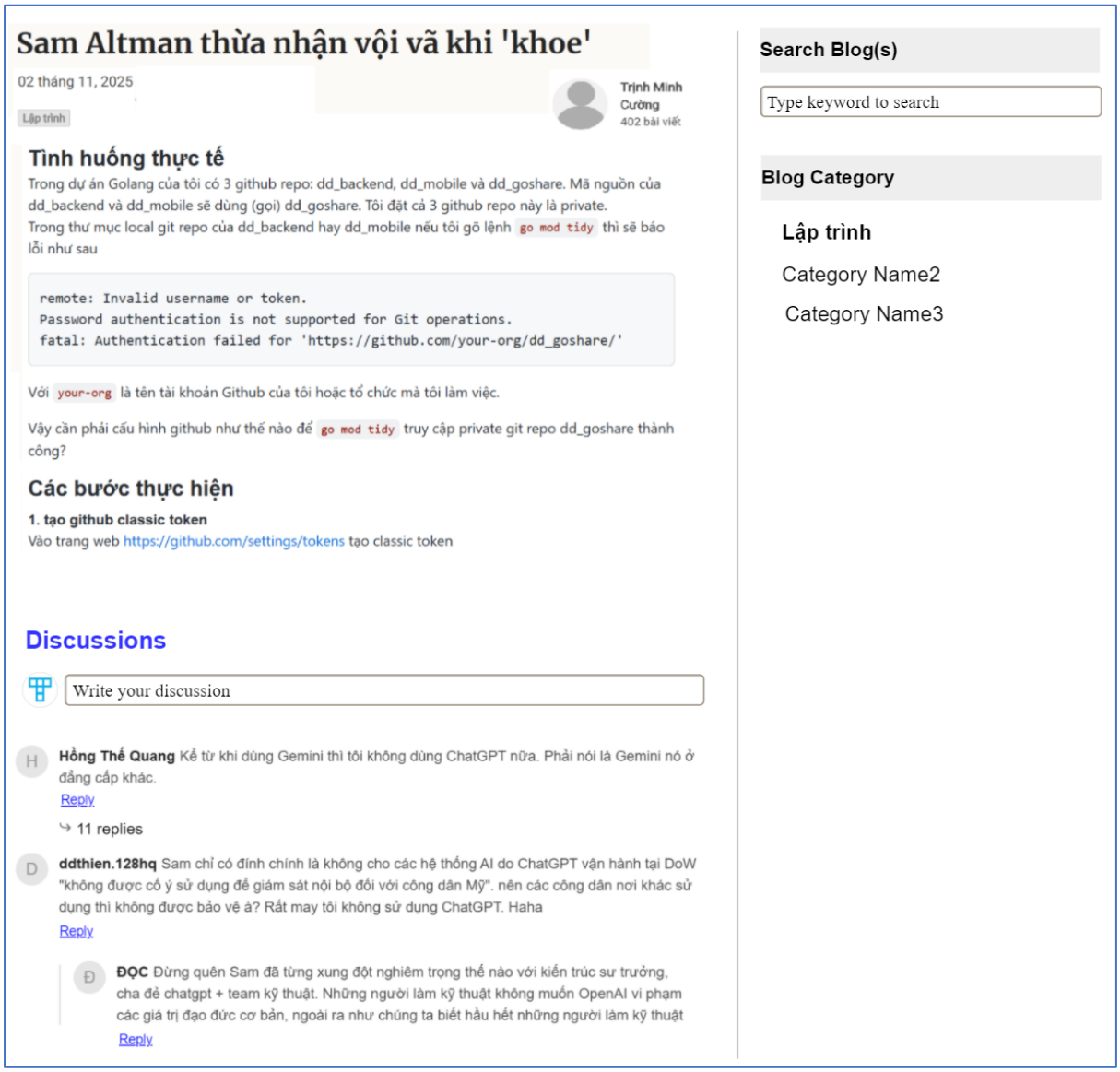
1.2. Public Courses



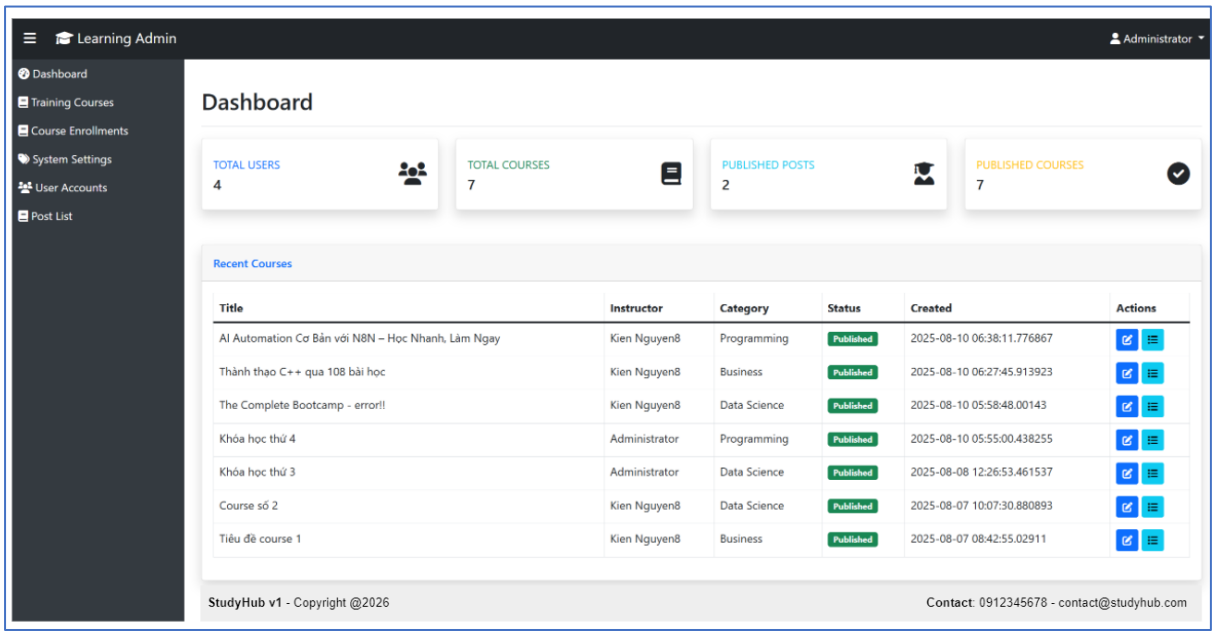
1.3. Blog List



1.4. Blog Details

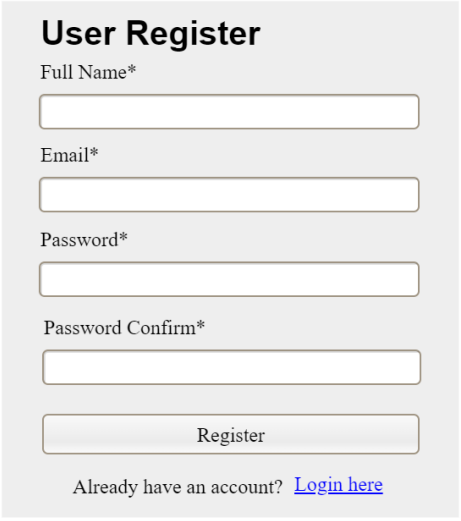


1.5. Admin Dashboard



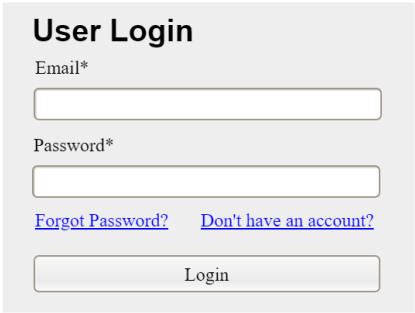
2. User Authentication

2.1. User Register



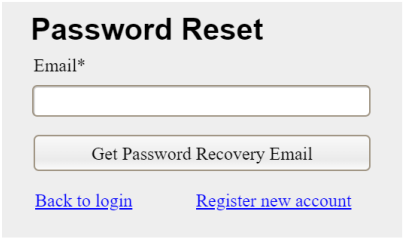
After successfully registering, a verification link would be sent to the registered email allowing user to click to verify his/her account for logging in.

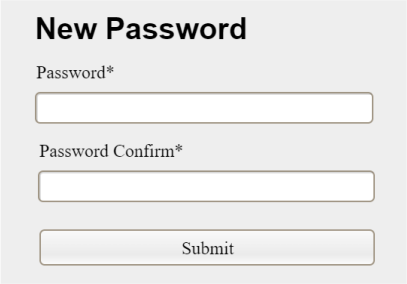
2.2. User Login



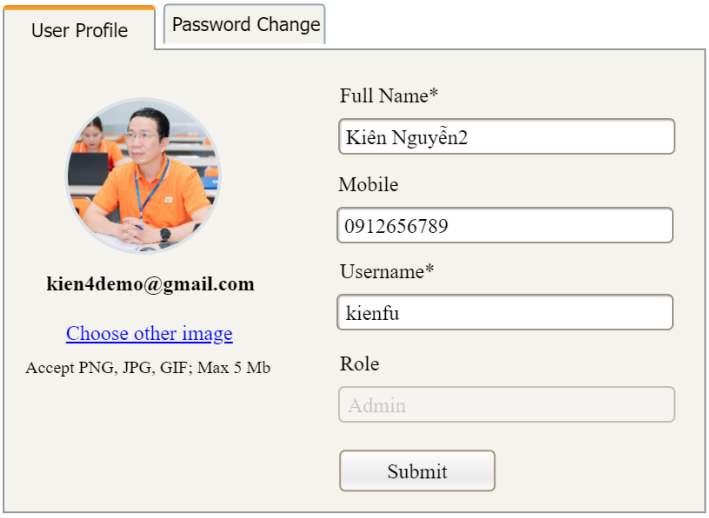
In case user logs in with an accout that the email has not been verified yet, he/she would get an according notification message with the option to resend the verifying link via email to verify his/her account.

2.3. Password Reset

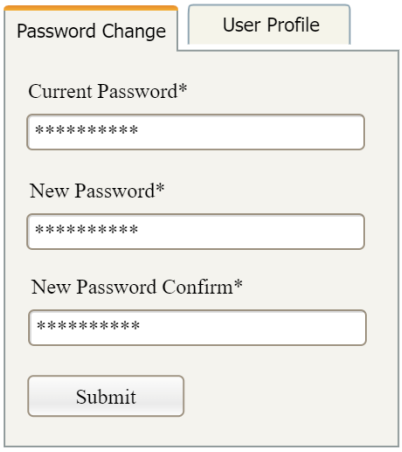




2.4. User Profile



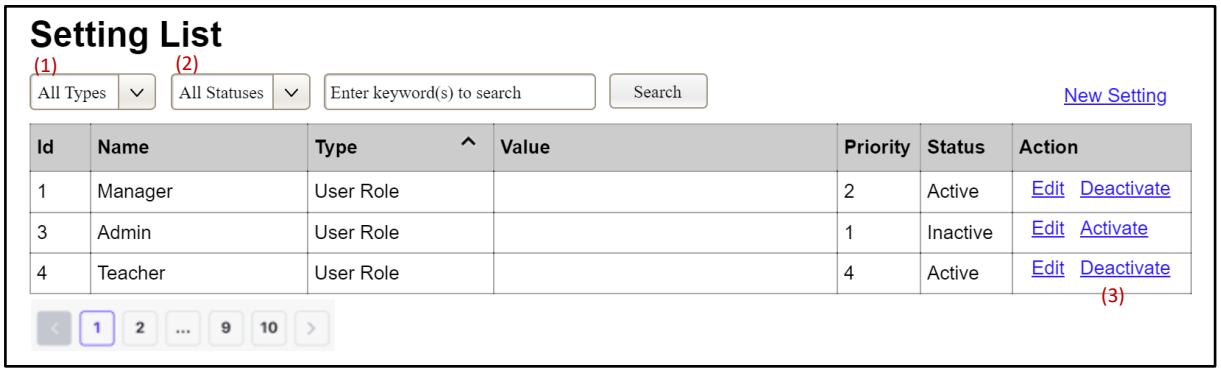
2.5. Password Change



3. System Administration

3.1. System Setting

a. Setting List



This screen allows the Administrator to:

- View Setting List: view list of current master data.
- Filter Setting List: filter master data by data types, statuses
- Search Settings: enter keyword(s) to search master data by their names or values
- Sort Setting List: sort master data list (ascending, descending) by clicking column headers

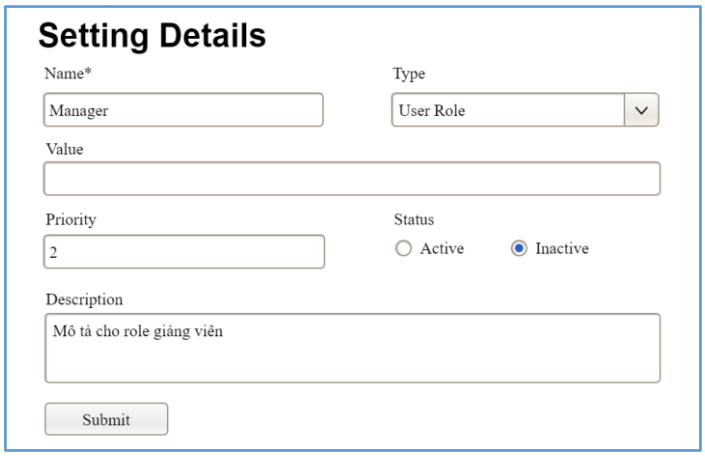
On the screen, s/he can also

- Activate/Deactivate Setting: change status of a specific inactive/active master data
- Choose to go to the Setting Details screens for adding new or updating an existing master data by clicking the New Setting or Edit link.

Field Description

Field Name	Description
(1)	Initial values: all the active setting names with null or blank type Hover the mouse to show the field name: “Setting Type”
(2)	Initial values: All Statuses, Active, Inactive (default value “All Status”) Hover the mouse to show the field name: “Setting Status”
(3)	The change-status action is Activate or Deactivate depending on the current status of the relevant setting (Inactive or Active, respectively).

b. Setting Details



This screen allows the Administrator to:

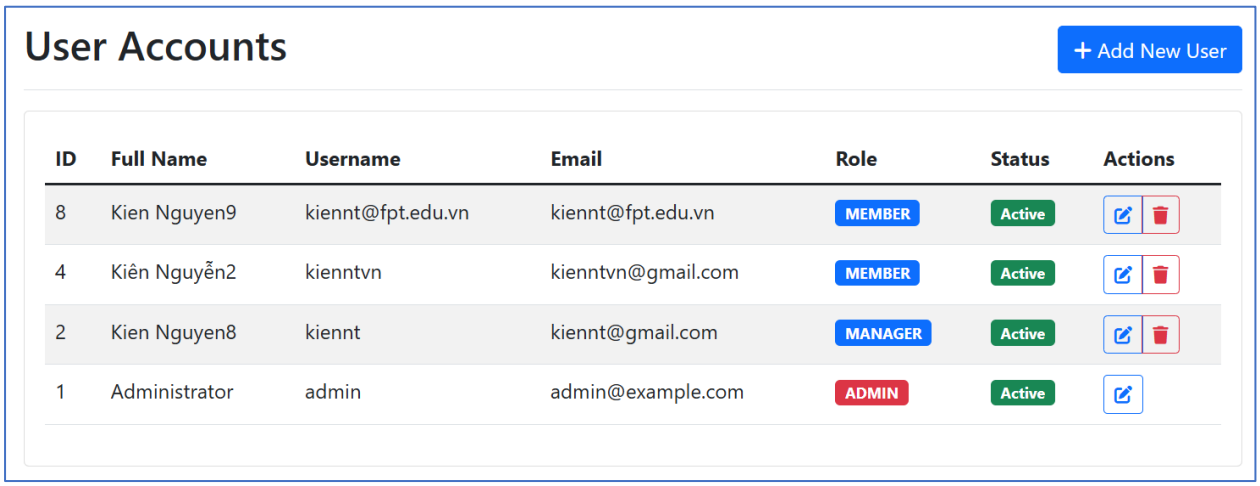
- Add New Setting: add new master data.
- Update Setting Details: update details of a specific master data

Field Description

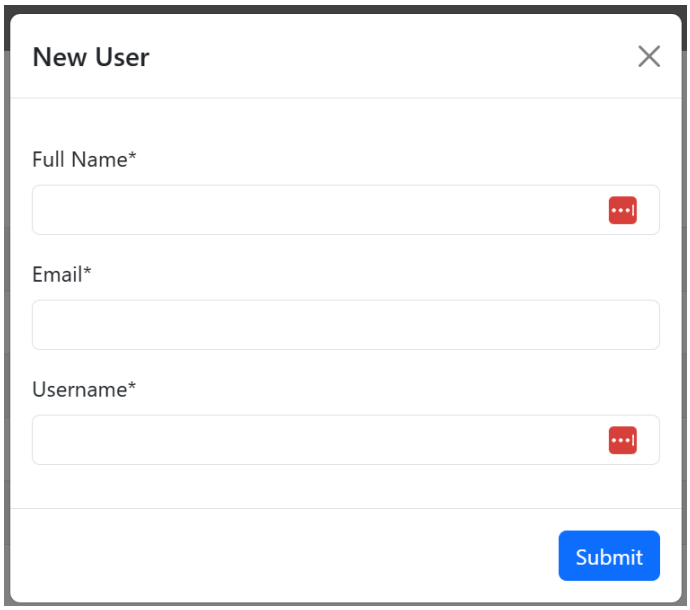
Field Name	Description
Name	Data type: non-digit string, max length of 20 characters Constraints: unique value in the selected type
Type	Initial data values: all active setting names (with null or blank type)
Value	Data type: any string, max length of 100 characters
Priority	Data type: a positive integer
Description	Data type: any string, max length of 200 characters

3.2 User Management

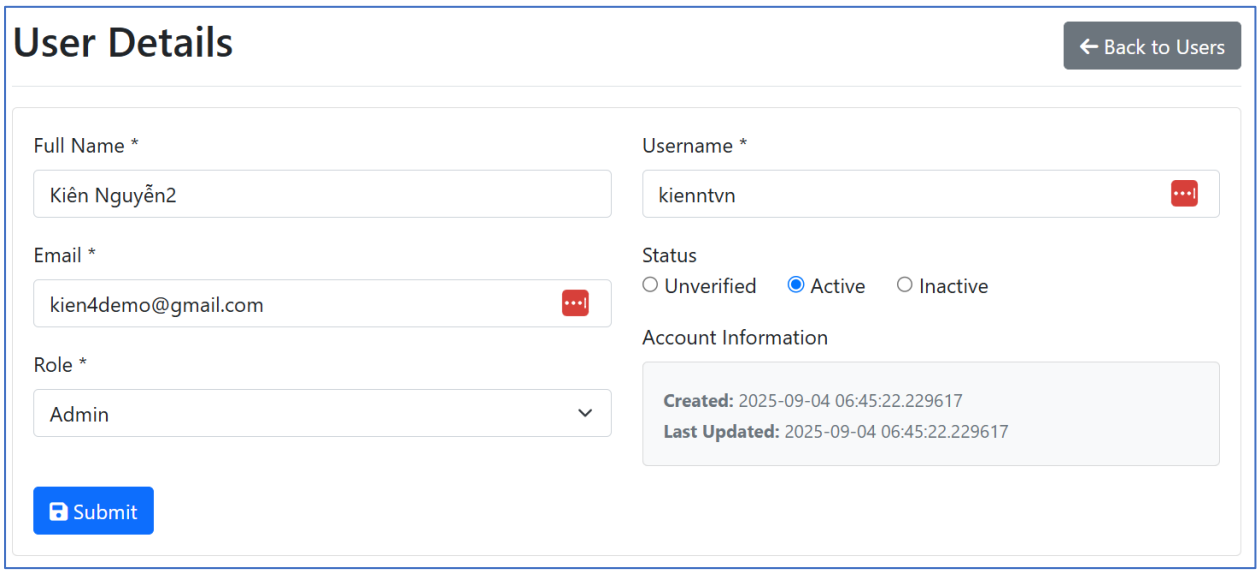
a. User List



b. New User



c. User Details



3.3. Post Management

3.3.1 Post List

Post List

All Categories | All Authors | Type keyword to search | Search | New Post

Id	Title	Category	Author	Comment	Status	Updated At	Actions
38	AI Automation Cơ Bản với N8N – Học Nhanh, Làm Ngay	Lập trình	Kien Nguyen	3	Hidden	30/09/2025 23:55:26	Edit Show
34	The Complete Bootcamp - error!!	Category Name2	Thuy TRan	0	Published	30/09/2025 13:09:15	Edit Hide
30	Khóa học thứ 4	Category Name2	Kien Nguyen	12	Draft	16/09/2025 11:49:04	Edit Show
28	Tiêu đề course 1	Category Name3	Tuan Minh	23	Published	01/10/2025 11:43:24	Edit Hide

3.3.2. Post Details

Post Details

Title*

Thumbnail Change

Category: Lập trình

Author: Author Name

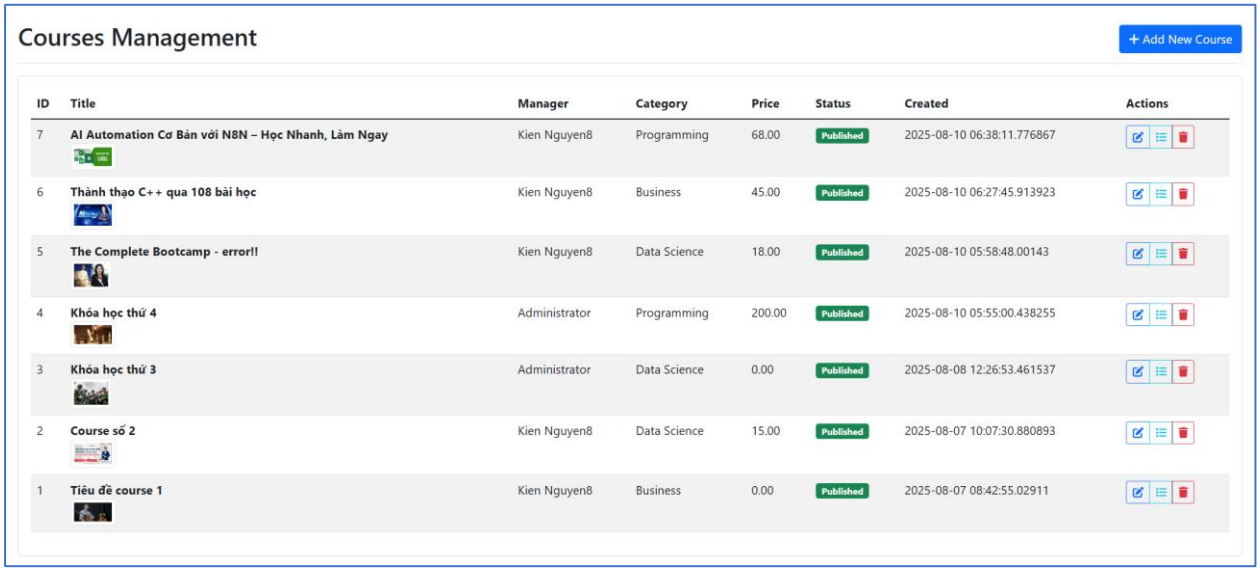
Status: Draft | Published | Hidden

Featured Post

Submit

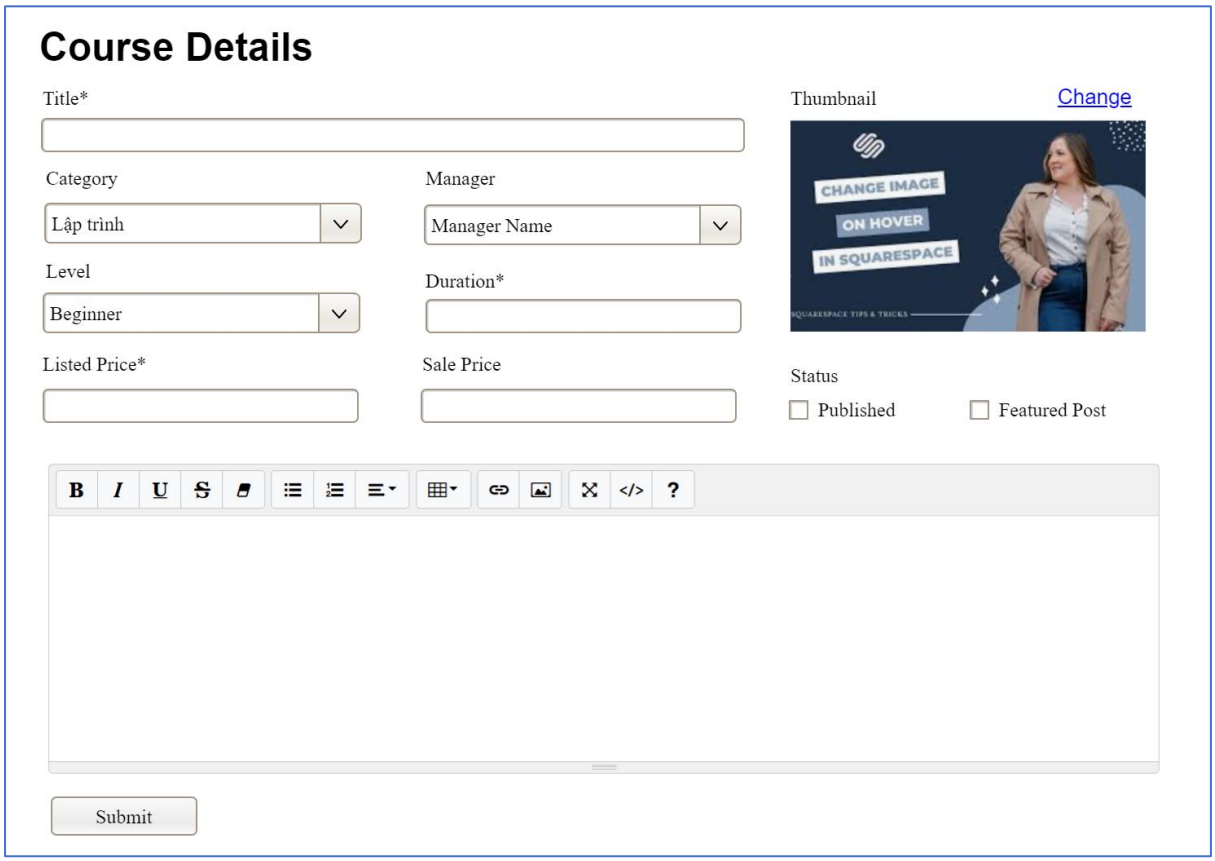
4. Course Management

4.1. Course List

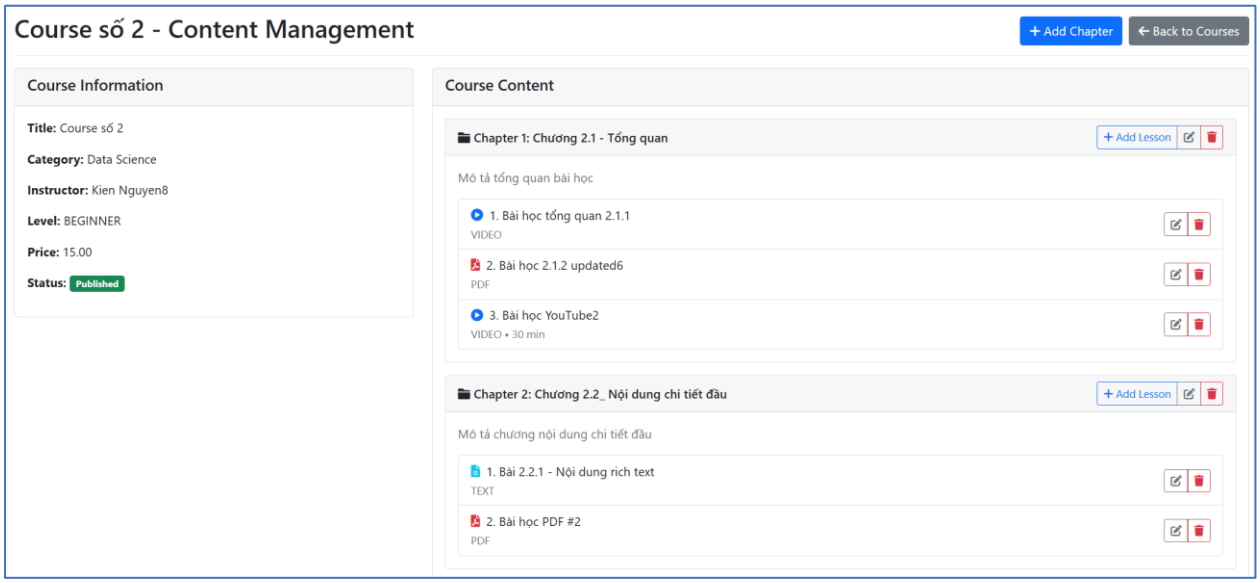


ID	Title	Manager	Category	Price	Status	Created	Actions
7	AI Automation Cơ Bản với N8N – Học Nhanh, Làm Ngay	Kien Nguyen8	Programming	68.00	Published	2025-08-10 06:38:11.776867	
6	Thành thạo C++ qua 108 bài học	Kien Nguyen8	Business	45.00	Published	2025-08-10 06:27:45.913923	
5	The Complete Bootcamp - error!!	Kien Nguyen8	Data Science	18.00	Published	2025-08-10 05:58:48.00143	
4	Khóa học thứ 4	Administrator	Programming	200.00	Published	2025-08-10 05:55:00.438255	
3	Khóa học thứ 3	Administrator	Data Science	0.00	Published	2025-08-08 12:26:53.461537	
2	Course số 2	Kien Nguyen8	Data Science	15.00	Published	2025-08-07 10:07:30.880893	
1	Tiêu đề course 1	Kien Nguyen8	Business	0.00	Published	2025-08-07 08:42:55.02911	

4.2. Course Details

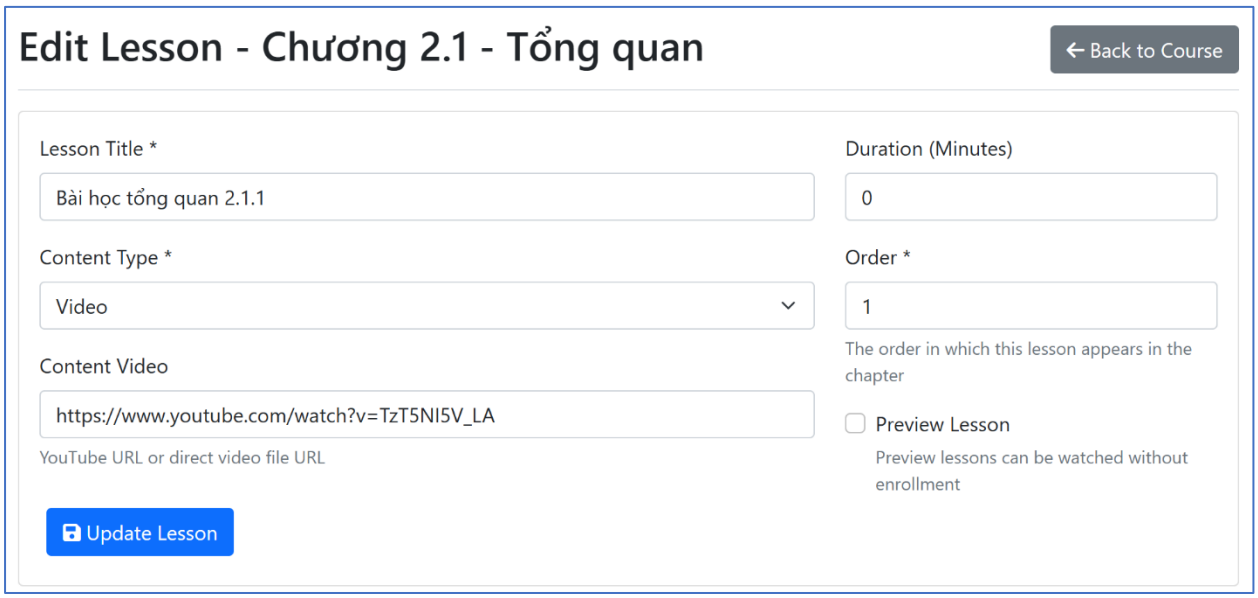


4.3. Course Content



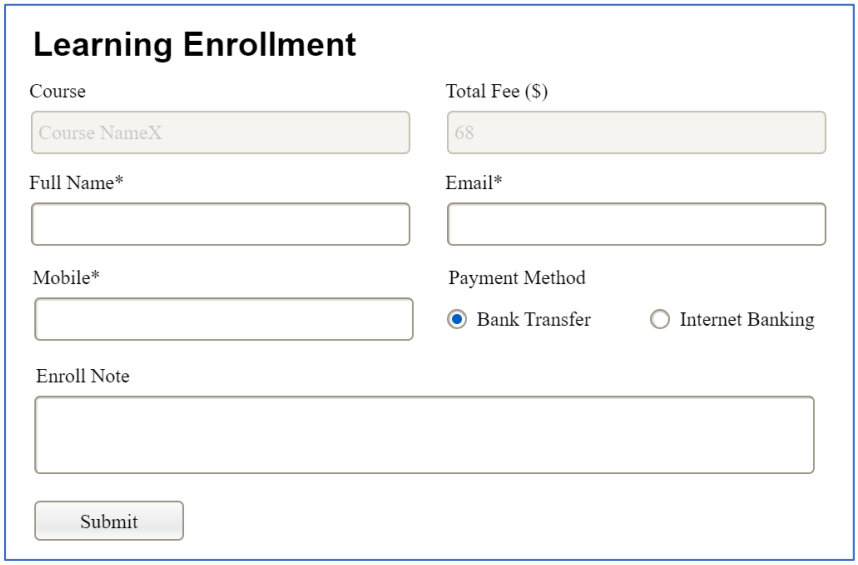
4.4. Chapter Details

4.5. Lesson Details



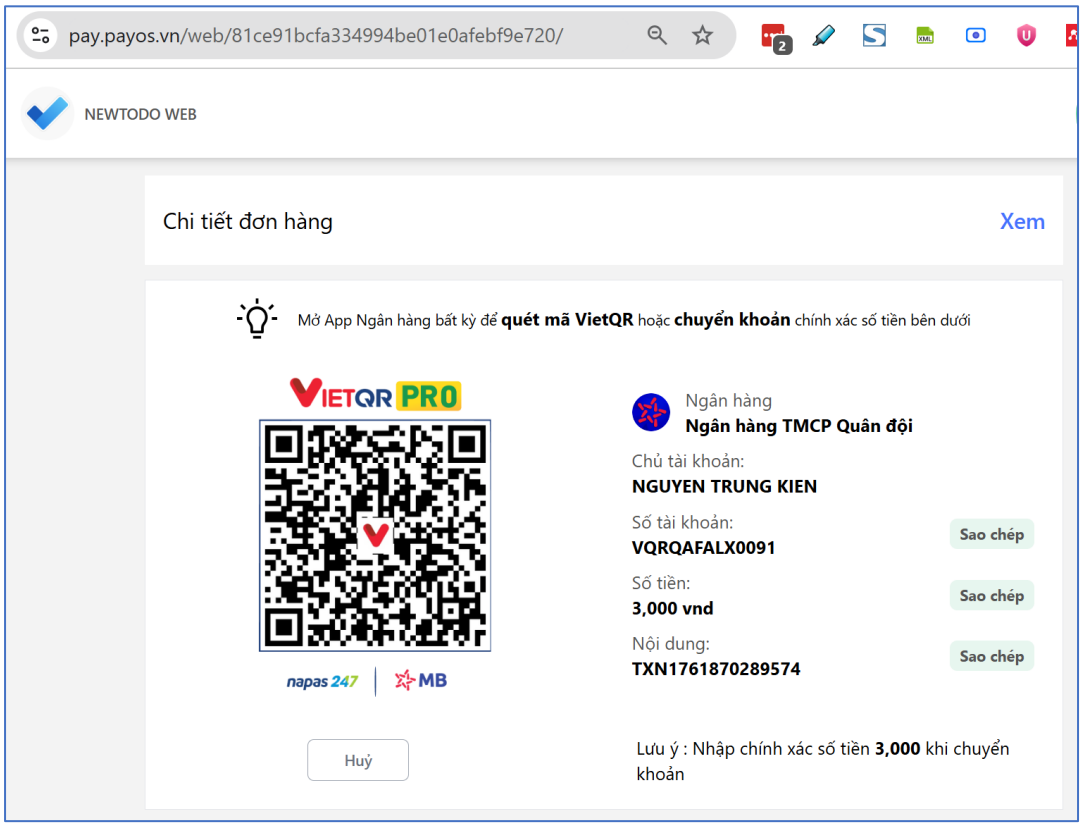
5. Course Enrollment

5.1. Learning Enroll

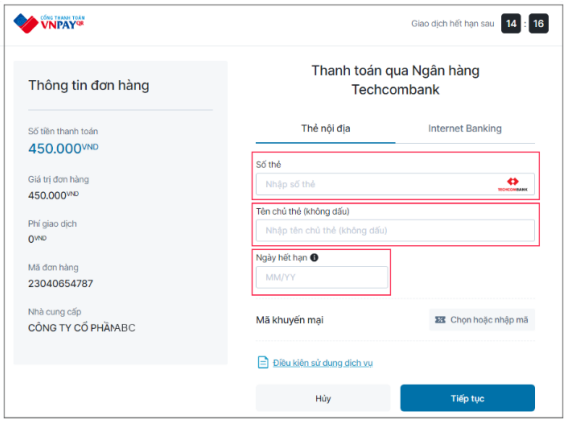


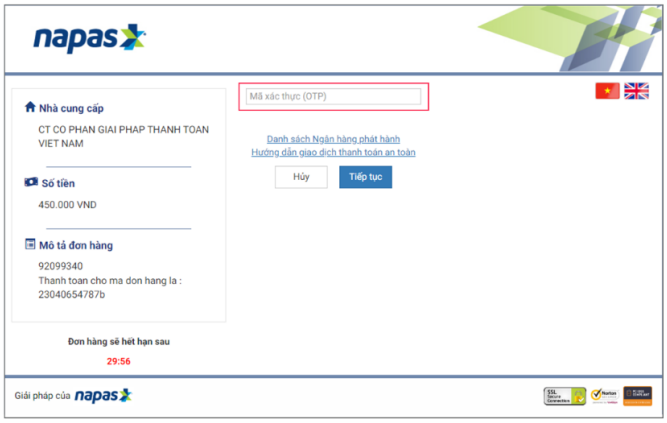
5.2. Online Payment

a. Bank Transfer (PayOS)



b. Internet Banking (VnPay)



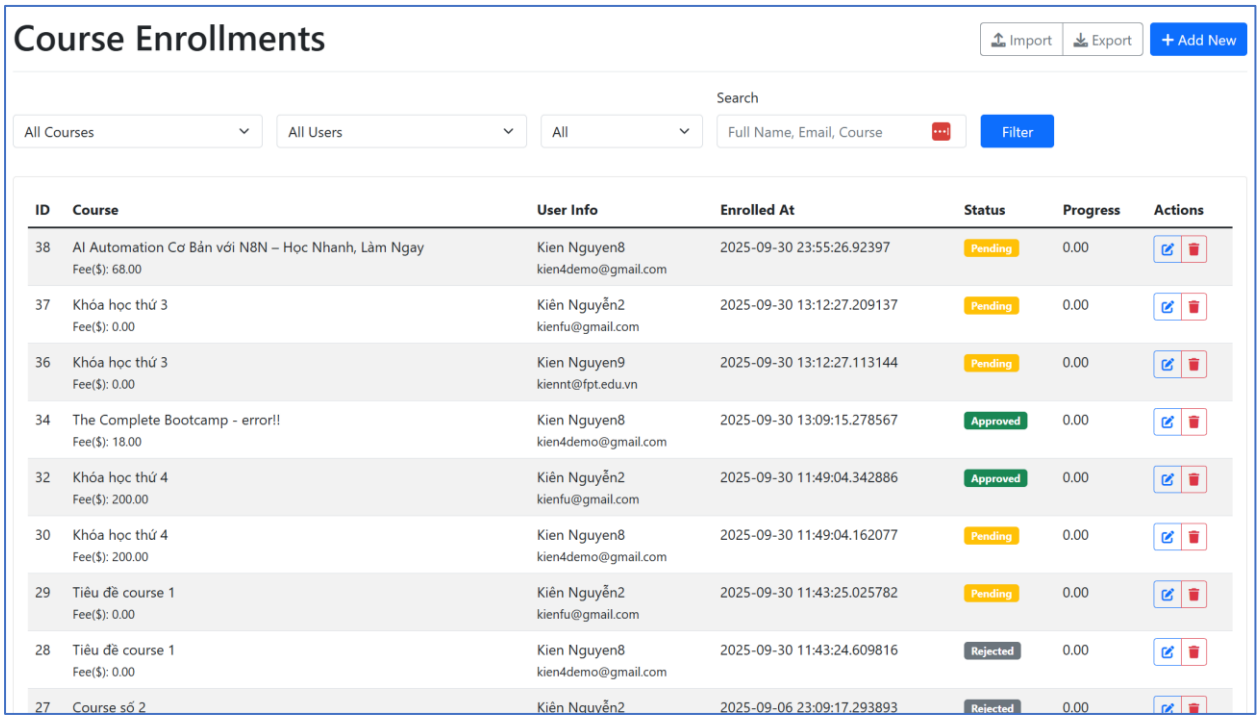


5.3. My Enrollments

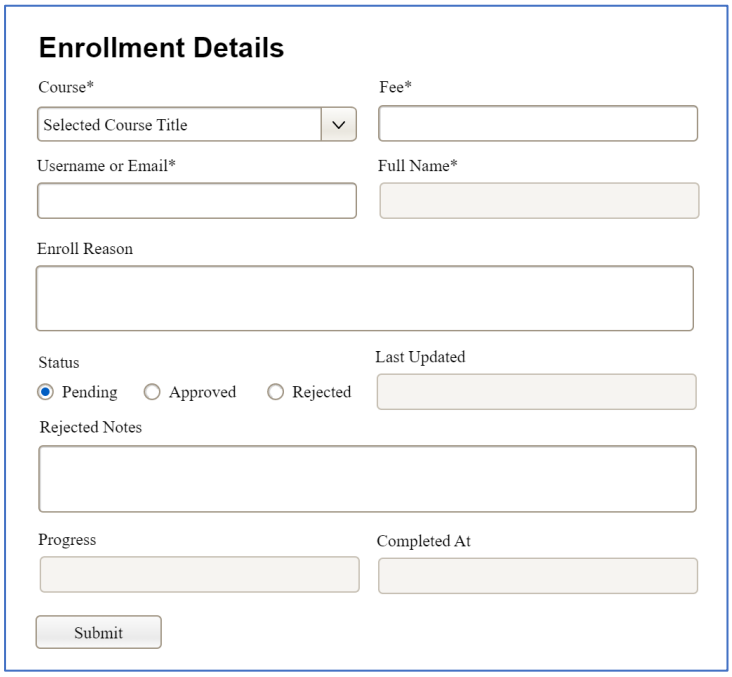
My Enrollments

Id	Course	User Info	Enrolled At	Fee($)	Status	Actions
38	AI Automation Cơ Bản với N8N – Học Nhanh, Làm Ngay	Kien Nguyen8 kien4demo@gmail.com	2025-09-30 23:55:26	68.00	Pending	Update
34	The Complete Bootcamp - error!!	Kien Nguyen8 kien4demo@gmail.com	2025-09-30 13:09:15	18.00	Paid	Access
30	Khóa học thứ 4	Kien Nguyen8 kien4demo@gmail.com	2025-09-30 11:49:04	200.00	Pending	Update
28	Tiêu đề course 1	Kien Nguyen8 kien4demo@gmail.com	2025-09-30 11:43:24	0	Cancelled	View

5.4. Enrollment List

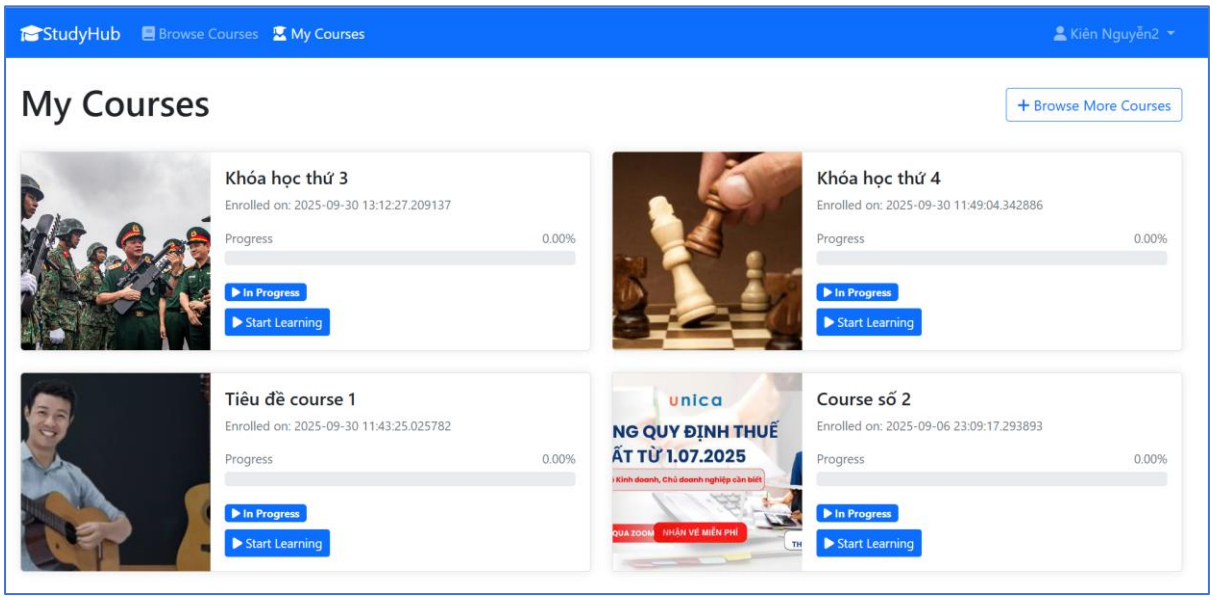


5.5. Enrollment Details

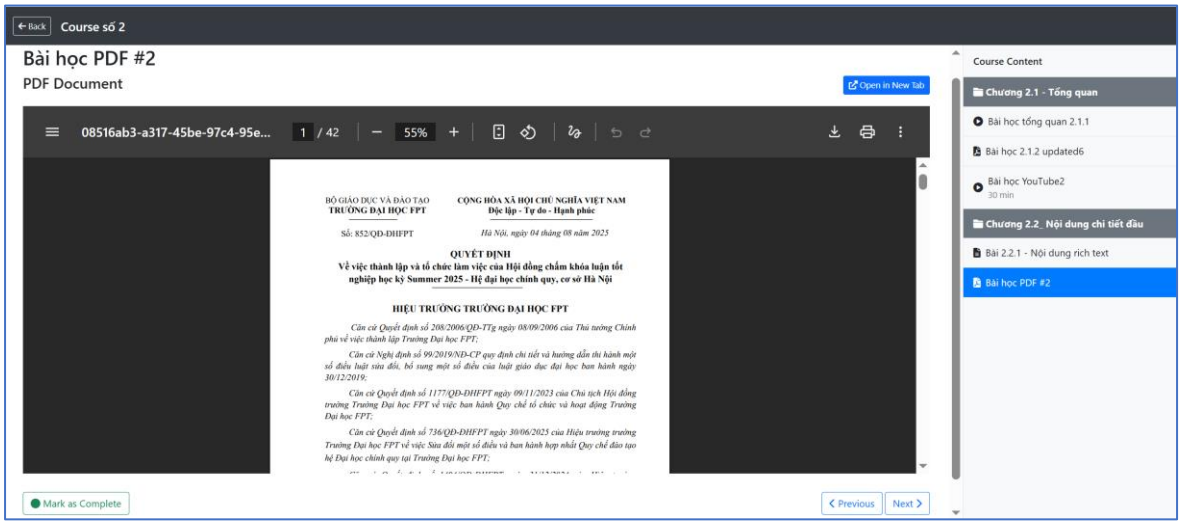


6. Course Access

6.1. My Courses

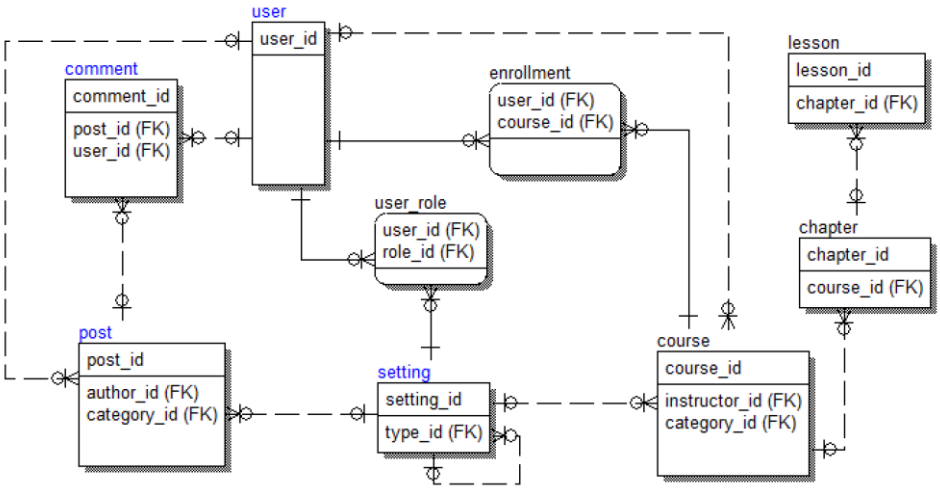


6.2. Lessons Viewer



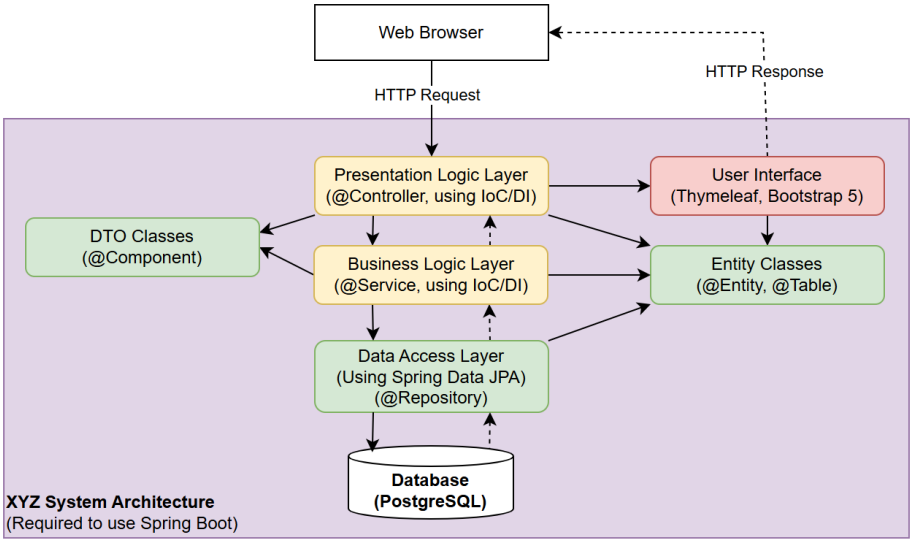
III. System Design

1. High Level Table Design



2. System Architecture

You are required to use the Spring Boot with details and the multi-layer architecture with specific library, framework or annotation as mentioned in the diagrams below



For the UI, you need to separate the header, footer, and sidebar into individual components so that they can be integrated and used together across pages.